6.1 (c) MODELLING FUNCTION & FLOW ASPECTS
- ACTIVITY DIAGRAM

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Activity diagrams

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UML activity diagrams can be used as supplementary or alternative graphic documentation to formulate the action sequences in use cases

Activity diagrams are often more easily readable than textual descriptions

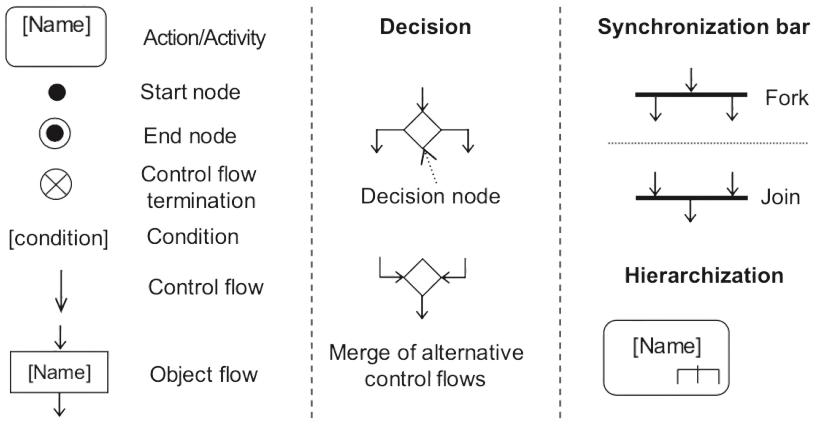
However, activity diagrams should not completely replace the textual description in a use case model

- the context of a use case is more understandable for the reader in textual form
- activity diagrams facilitate the reading of complicated sections of the action sequences of a use case
- textual description and activity diagrams must be synchronized with each other

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Activity Diagram - Syntax & Semantics



Important

More info on AD elements

https://sparxsystems.com/resources/tutorials/uml2/activity-diagram.html

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In-Class Activity 1

- **Watch the "All About UML Activity Diagrams" video at** https://www.youtube.com/watch?v=Wf_xlagfHmg
- **Draw mind-map to summarize your understanding on 5W1H important points of activity diagram**
- *** 5W1H = What, When, Who, Where, Which & How**

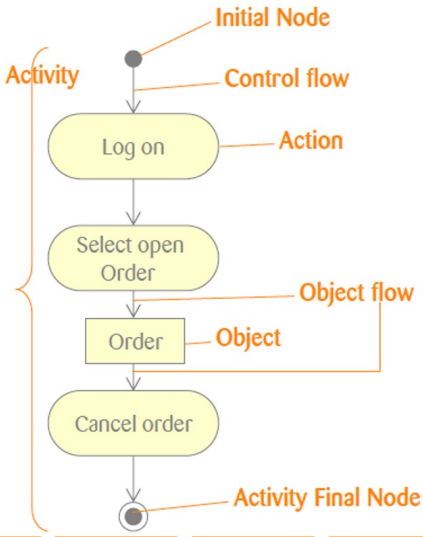
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Activity diagrams – Basic elements

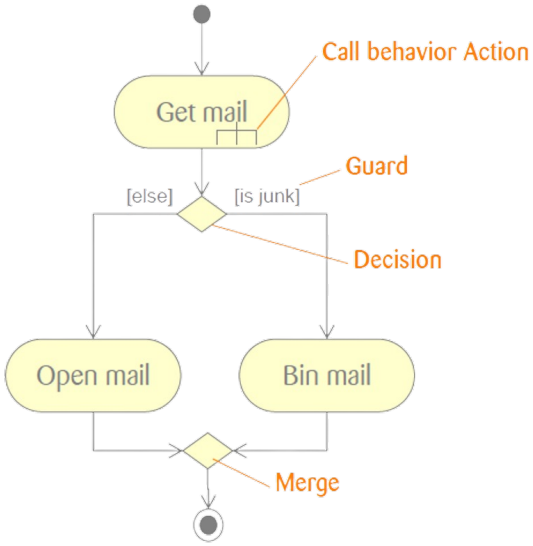
- Token-flow (Petri net) semantic
- The activity starts with tokens at the flows of the initial node
- Control flows pass on tokens
- Actions are executed when all ingoing flows provide a token.
  After execution actions provide tokens at all outgoing flows.
- Object flows pass on objects
- The activity final node terminates all execution inside the activity



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Activity diagrams – Alternative flows

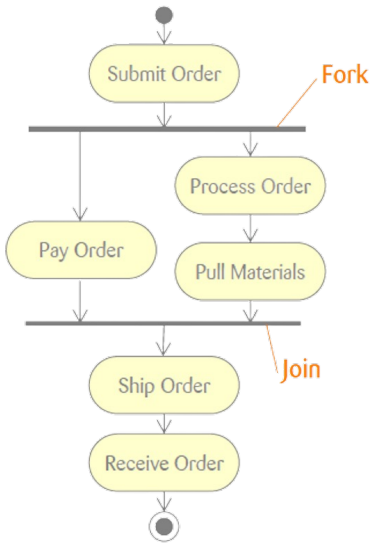
- Decision nodes pass on tokens to outgoing edges. Which edge is traversed depends on the guards of the edges.
- The [else] guard is predefined in UML.
- Merge nodes brings alternative flows together.
  Tokens, arriving at an incoming edge are passed on to the single outgoing edge.
- Merge and decision nodes can be combined in a single node.
- Call behavior Actions invoke other activities which are defined in separate diagrams (decomposition).



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Activity diagrams – Concurrent flows

- Fork nodes split the flow into multiple concurrent flows.
  Incoming tokens are duplicated for all outgoing flows.
- Join nodes synchronize multiple concurrent flows.
  A token is passed on when all incoming edges provide a token.
- Join and fork nodes can be combined in a single node.



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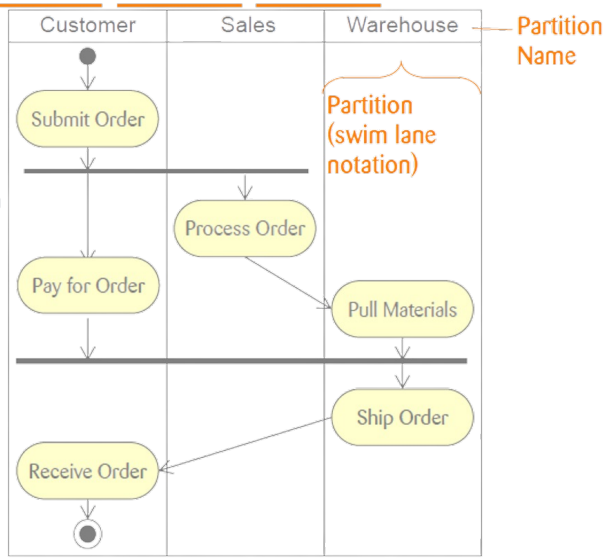
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Activity diagrams – Assignment of responsibilities

- Partitions group actions with common characteristics.
  Typically used to define responsibilities.

Examples of partitions

- Use case actors and system
- organizational units of a company

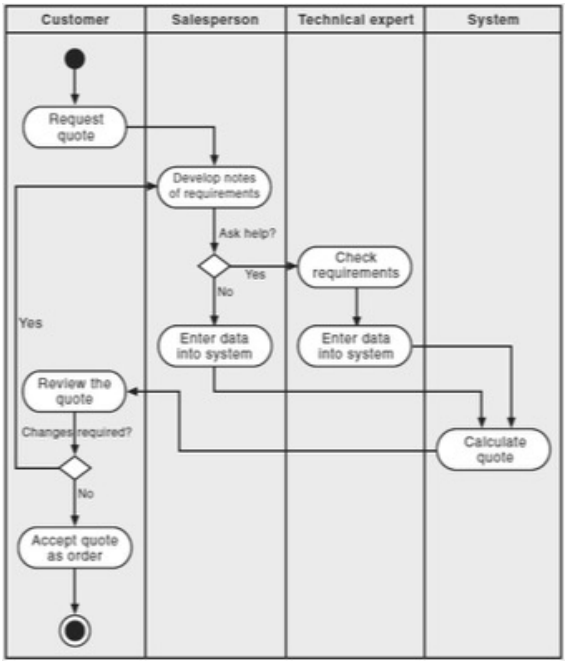


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In-Class Activity 2

Interpret the workflow that represents a customer requesting a quote from a salesperson for a customer ordering system based on activity diagram in the Figure.



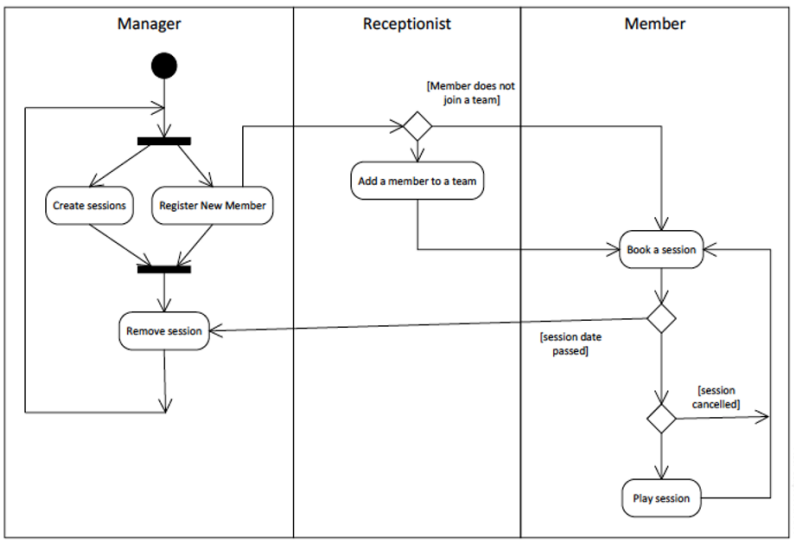
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In-Class Activity 3

Explain the activity diagram of the Badminton club activities and processes based on activity diagram in the Figure.



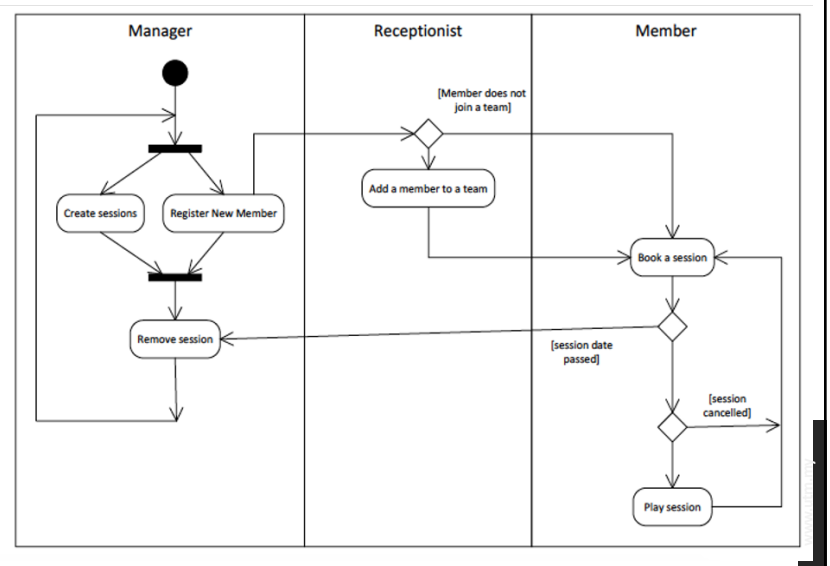
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In-Class Activity 4

Explain the activity diagram of the Badminton club activities and processes based on activity diagram in the Figure.



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THANK YOU

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